Supplementary Material: Governance Delay and Institutional Feedback

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This supplement gives the equations, proofs, numerical protocols and evidence supporting the main article. Local nonlinear Hopf hypotheses and protective review/viability inequalities are distinguished from finite-dimensional fold certificates, floating periodic-orbit evidence and observational event-clock constraints. S1–S12 contain the core methods and model comparisons; S13–S16 give the nonlinear normal-form derivation, all-cadences and stock-floor details, connected-branch evidence and institutional-event protocol. S17–S20 supply complete equilibrium spectra, the complete protective basin, full Fourier residual bounds and quantitative field tests.

Notation. We use the symbols of the main article. The state variables are the stock N, the institutional memory Z, and the deployed effort E. The mobilising and protective feedback channels carry linearised coefficients (C_E, C_Z) with subscripted variants for the two channels. The delay τ is deployment latency; T_r is periodic review cadence. Governance capital, when introduced in S4.4, is K_{cap}, not Z. The candidate parameter sets A and B and the four-state working core are as in the main article. The following terms recur throughout and are glossed here on first use: an *interval Newton method* is a validated-root enclosure technique that returns a rigorous box containing any root of a system in a declared interval; a *Krawczyk inclusion* is a validated interval enclosure constructed from the Krawczyk operator; a *Moore–Spence system* is the standard augmented system for fold computation; a *radii polynomial* is a constructive validation tool for infinite-dimensional problems; a *Floquet multiplier* is an eigenvalue of the monodromy map for a periodic orbit; a *Hopf enclosure* is an interval that contains the bifurcation delay with rigorous outward rounding; and *collocation* denotes a Fourier Galerkin discretisation of the periodic-orbit boundary-value problem.

S1. Interval Hopf Enclosures

This section gives the validated enclosures for the delays at which the gated Candidate A equilibrium changes linear stability. The enclosures underwrite the qualitative Hopf statement of the main article; they do not, by themselves, amount to a validated Hopf bifurcation theorem. That distinction is preserved as the scope paragraph S1.3 below.

Definition S1.1 (Cubic and phase relation). With $P(\lambda) = (\lambda - A_N)(\lambda + d)(\lambda - C_E)$ and $L(\lambda) = B_E(\lambda - A_N) + A_E B_N$, the modulus condition of Theorem 4.1 of the main article gives the explicit cubic

$$H(x) = (x + A_N^2)(x + d^2)(x + C_E^2) - C_Z^2\big[B_E^2 x + (A_E B_N - A_N B_E)^2\big],$$

and at the interior equilibrium of the gated core the filter identity $A_E B_N - A_N B_E = 0$ reduces it to $H(x) = (x + A_N^2)(x + d^2)(x + C_E^2) - C_Z^2 B_E^2 x$. The delay is the interval evaluation of the phase relation $\tau(\omega) = (-\arg(P(i\omega)/(C_Z L(i\omega))) + 2\pi k)/\omega$ at a certified positive root of H, not a root of an argument formula.

Proposition S1.2 (Certified enclosures, gated Candidate A). Outward endpoint-sign and monotonic-derivative tests on the interval-enclosed coefficients certify simple positive roots in $x = \omega^2$, and branch-safe interval evaluation of the phase relation gives

$$\tau_- \in [3.6661490142739,\ 3.6661490142743], \qquad \tau_+ \in [150.3584773101408,\ 150.3584773101421],$$

with upper-interval width of order 10^{-12} yr. The certificate is conditional on the stated interval coefficient and phase-evaluation pipeline, including correct handling of the argument branch; the deposited pipeline implements the outward-rounded coefficient/equilibrium/phase construction with interval transcendentals and reproduces the displayed intervals exactly on re-execution. A full independent record must state the interval library, working precision, rounding mode, parameter and equilibrium enclosures, coefficient intervals, and the branch-safe implementation of the argument function — implementation details that are part of an interval certificate, not merely numerical-method metadata.

Scope (S1.3). Scalar frequency/phase enclosures alone establish the local imaginary spectrum. S13 additionally verifies the classical nonlinear RFDE Hopf hypotheses for gated A and ungated B using

explicit second/third derivatives, interval simplicity/transversality checks, nonzero cubic coefficients and separation from all other imaginary frequencies. A constructive radii-polynomial branch neighbourhood and both continuum fold lifts remain separate. No numerical delay threshold is calibrated to a named fishery.

S2. Collocation Formulation and the Small-Branch Computation

This section reports the collocation discretisation used to continue the small-amplitude periodic orbit branch away from the lower Hopf crossing. The residuals reported here are nominal re-execution-verified values, not validated interval bounds; the gap between nominal computation and a validated fold certificate is documented in S3.

Definition S2.1 (Phase-fixed collocation map). With $m = 64$ Fourier nodes, $Y \in \mathbb{R}^{192}$ the coefficients of the three state variables, and $T > 0$ the unknown period, the phase-fixed collocation map is $F : \mathbb{R}^{192} \times \mathbb{R} \to \mathbb{R}^{193}$, with the phase equation among the 193 equations, removing the time-translation degeneracy; $D_{(Y,T)}F$ is 193×193. The map is a finite-dimensional collocation map; no continuous-DDE truncation bound is implied by its residual.

Proposition S2.2 (Small-branch continuation). A Newton–Levenberg–Marquardt solver, initialised from the Hopf normal-form predictor at $\tau = \tau_- + 0.05$ and continued with a $\sqrt{\tau - \tau_-}$ predictor, finds one approximate periodic-orbit solution at each sampled $\tau \in [3.716, 5.58667]$ with discretised residual $\|F\|_2 \le 6 \times 10^{-14}$; the peak-to-peak amplitude of N grows from 1.10 to 21.80 and the period from 250.0 to 313.76 yr. The solver fails at $\tau = 5.590$ under the stated budget with residual 2.8×10^{-6}: a solver-success/failure bracket, not a nonexistence result. The fold pipeline at $m = 64/96/128$ gives nominal fold values $\tau_f = 5.587236198690$, 5.587236198663, and 5.587236198663, in cross-resolution agreement to 2.7×10^{-11} — nominal values, re-execution-verified at the nominal level, not certificates.

Proposition S2.3 (Last collocation point). At $\tau = 5.586666666666667$ the computed orbit satisfies $\|F\|_2 = 5.43 \times 10^{-14}$, $\|J^{-1}F\|_2 = 5.69 \times 10^{-12}$, with $\sigma_{\min}(J) = 4.54 \times 10^{-7}$ and $\mathrm{cond}_2(J) = 1.25 \times 10^{7}$; the effort gate and filter floor are inactive ($E = 7.93 \ll E_{\max}$, $N \ge 72.2$). $\|J^{-1}F\|_2$ is the norm of a linearised Newton correction for the finite-dimensional collocation map; it is not a rigorous error bound for the continuous equation or for the fold location. The data do not establish uniqueness of the collocation zero at each τ, and solver failure at 5.590 does not exclude another zero or family there.

Remark S2.4 (Companion Floquet evidence). Independent shooting/Floquet calculations of the companion core give a real nontrivial multiplier changing from approximately 1.0514 at $\tau = 5.584$ to 0.99898 at $\tau = 5.587$ on the corresponding small branch, consistent with a candidate simple turning point. This is not an interval certificate, and the parameter-mismatch caution applies (5.58667 and the fold-quoted 5.587 are not the same parameter). For computational methodology the multiplier drift establishes that the collocation branch behaves locally like a fold candidate; it does not establish that the candidate survives validation.

S3. Discrete Fold Certificates and Continuum Requirements

For the phase-fixed $m = 64$ collocation map $F(w, \tau)$, with $w = (Y, T) \in \mathbb{R}^{193}$, the Moore–Spence system is

$$\mathcal{M}(w, \tau, v) = (F(w, \tau), D_w F(w, \tau)v, \ell^\top v - 1) = 0.$$

It has 387 equations and unknowns. An interval Krawczyk inclusion establishes a unique zero in the finite-dimensional box. The archived lower and upper fold certificates respectively enclose

$$\tau_f \in [5.587236198689235, 5.587236198690484],$$

$$\tau_{f2} \in [64.40232720336789, 64.40232720337167].$$

The left-nullvector contractions give the lower-fold constants $w^\top F_\tau \in [0.3132662, 0.3148219]$ and $w^\top D^2F[v, v] \in [5.6923, 5.8943] \times 10^{-5}$, and the upper-fold constants $w^\top F_\tau \in [0.2579355, 0.2607498]$ and $w^\top D^2F[v, v] \in [-5.88663, -5.87702] \times 10^{-4}$. Thus the free-delay finite-dimensional system and its

discrete fold nondegeneracy are certified. The exact certificate JSONs are preserved in the package's inherited evidence directory; these are archived certificates, not newly rerun fold pipelines.

A continuum result requires a bordered infinite-dimensional fold construction with controlled off-grid/truncation errors and continuum nondegeneracy. The nonlinear tail of $f(U, SU)$ does not vanish when U has finite Fourier support: softplus and rational terms generate infinitely many modes. Neither a small floating off-grid residual nor a periodic-orbit certificate without the bordered fold equations supplies this lift. S19 adds interval bounds for the whole infinite orbit/kernel residual series. The continuous-delay saddle-node theorem still requires the bordered inverse, contraction and nondegeneracy proof. No fold statement transfers to a different turnover, tether, stage-structured or spatial model without fresh analysis.

S4. The Stress-Test Frame

This section reports the interpretive frame that disciplines the transfer of any bifurcation reading to a sustainability claim. The frame is built to prevent overreach: each bifurcation is read inside its declared response-sign class, and no result for one class is generalised to another.

Definition S4.1 (Response-sign taxonomy). The interpretive frame of the two channels is a three-hypothesis response-sign taxonomy:

- (H1) Scarcity-amplifying extraction: $\partial \mathcal{G}_{\mathrm{H1}}/\partial Z > 0$ over the relevant range.
- (H2) Protective restraint or restoration: the reversed or restoration-including effect on extraction.
- (H3) Inertia, capture, or state-dependent response.

The frame's discipline is preserved verbatim: no result for H1 is generalised to H2 or H3. H1's named instantiation is the mobilising family of Section 5 of the main article; H2's named instantiation is the protective channel of Section 6; H3 belongs to the institutional-implementation family treated elsewhere. The model must state whether E is an endogenous industry response, a legal quota-utilisation state, or an actual institutional control; these interpretations are not interchangeable, and a model-specific bifurcation threshold is not a universal policy threshold. Empirical clocks require separately dated endpoints. Hocherman, Trop and Ghermandi (2025) synthesise 101 heterogeneous studies, not an identified 2–13-year implementation-lag distribution. The within-stock event protocol of S16 preserves advice, action and compliance distinctions and does not identify the constant model delay from calendar differences.

Definition S4.2 (Liquidation mechanism types). The phrase "liquidation" has three distinct physical meanings:

- (i) Standing-stock culling: present extraction removes reproductive stock directly.
- (ii) Recruitment suppression: present use prevents future recruits without immediate adult removal.
- (iii) Weak viability coupling: limited or indirect effect on reproduction.

Recruitment suppression modifies or diverts a recruitment flux and is one typed-flux incidence family with culling and weak coupling; its named treatment in the main article is the recruitment channel of the two-channel stock law (Section 2.3). A diagnostic label never determines physical destination: material routing is determined by the typed physical module, not by whether a flow exceeds a benchmark.

Definition S4.3 (Variant registry). Nine numerical-programme variants are archived: ungated, gated, hybrid-effort, four-state support-pool, two-channel liquidation, stage-structured, sampled-review, thermodynamic-tether, and unified-core. Each row is a registered obligation where no verified artifact exists: it requires source equations, parameters, scripts or software settings, and outputs before any numerical claim. The tabulation discipline is: governing equations and state/control interpretation; parameter units and identifiability status; equilibrium and boundary conditions; numerical method, mesh, time-step, event handling, and horizon; branch identity and uncertainty; and the status label (analytical, independently reproduced numerical, conjectural, or superseded). Distinct claim types are kept distinct: a persistence boundary, a fold, an SNPO-confirmed fold, and an initial-condition basin observation are different claims, and no fold is called SNPO-confirmed without branch-specific multiplier and transversality evidence. Two discipline remarks frame every row: a frozen pool is a formal limiting case, not a default physical assertion; and gating, damping, or saturation is not accepted merely because it preserves a desired bifurcation — it must have a constitutive interpretation and an independently stated parameter role (the multiplicative gate of the gated core does: it is the effort ceiling's hard saturation). The safety-relevance ladder governs interpretation: a local bifurcation in a reduced model

is not automatically a sustainability transition — the three levels are existence of the bifurcation in a specified model, persistence of the organising structure under physically admissible coupling, and safety relevance, measured by the intersection of an attractor, basin, or uncertainty tube with the declared unsafe set, where basin membership and not only attractor location matters in bistable systems.

S4.4 Thermodynamic tether: positivity, exact equilibrium criterion and an explicit Allee example

For physical effort with an energetic cost, introduce governance capital $C = K_{\mathrm{cap}}$, distinct from carrying capacity K:

$$\dot{C} = qEN - C(\delta_K + c_E E),$$
$$\dot{E} = (1 - E/E_{\max})(1 - e^{-C/K_0})\mathcal{B}(E, Z_\tau) - \mu_E E,$$

where $\mathcal{B}(E, Z) = \eta E(Z - E/E_{\max}) + \delta_0 Z/(Z_{\mathrm{ref}} + Z)$ at unit deficit reference, and all auxiliary coefficients are positive. The integrating-factor formula for C and nonnegative input qEN prove $C(t) \geq 0$. The effort interval $[0, E_{\max}]$ is invariant. Both the exponential gate and $C/(C + K_0)$ are regular on this physical capital region; a negative-capital pole is not a physical-state mechanism.

Proposition S4.1 (Exact positive-equilibrium criterion). Put $E_b = \min\{E_{\max}, r/q\}$, $N^*(E) = K(1 - qE/r)$, $Z^*(E) = \delta$, and

$$C^*(E) = \frac{qEN^*(E)}{\delta_K + c_E E}, \quad G(E) = \frac{(1 - E/E_{\max})(1 - e^{-C^*(E)/K_0})\mathcal{B}(E, \delta)}{E}.$$

Extend G continuously to $[0, E_b]$. Then

$$G(0) = \alpha = \frac{qK\delta_0\delta}{K_0\delta_K(Z_{\mathrm{ref}} + \delta)}, \qquad G(E_b) = 0.$$

For $\mu_E > 0$, positive interior equilibria are exactly the roots $G(E) = \mu_E$. In particular $\mu_E < \alpha$ is sufficient, not necessary, for existence. Define $\mu_{\max} = \max_{[0,E_b]} G$. No positive equilibrium exists above $\mu_{\max}$. Every $0 < \mu_E < \mu_{\max}$ has at least one; if $\alpha < \mu_E < \mu_{\max}$, there are at least two. An interior nondegenerate maximum ($G' = 0$, $G'' < 0$) is a fold of the equilibrium set. A dynamical saddle-node theorem additionally requires the corresponding RFDE spectral nondegeneracy.

Proof. Steady stock and memory equations give N^*, Z^*; steady capital gives C^*. The remaining equation is $E[G(E) - \mu_E] = 0$. The endpoint at zero follows by $1 - e^{-C/K_0} = C/K_0 + O(C^2)$. At E_b, either the effort gate or the yield/capital factor vanishes. Compactness and the intermediate value theorem give the stated root counts. The fold follows from the quadratic expansion at a strict nondegenerate maximum. Positive-equilibrium full-delay stability does not follow from root ordering alone.

The zero-effort equilibrium is $(K, \delta, 0, 0)$. Its characteristic polynomial factors into $(\lambda + r)(\lambda + d)$ and

$$\lambda^2 + (\mu_E + \delta_K)\lambda + \delta_K(\mu_E - \alpha).$$

The delayed-memory derivative in the effort row vanishes there. Consequently this rest is stable for $\mu_E > \alpha$ independently of delay, and unstable for $\mu_E < \alpha$.

Proposition S4.2 (Extinction-face scope). On $N = 0$, capital and effort decay to zero, yielding the rest $(0, \delta, 0, 0)$. In the full system this rest has stock eigenvalue $+r$ and is unstable to inward stock perturbations. Attraction on the invariant extinction face is not attraction from positive-stock initial conditions.

An explicit interval-checked Allee example. Use the Candidate A ecological/memory/bracket coefficients and auxiliary values $\delta_K = 0.1$, $c_E = 0.01$, $K_0 = 0.1$, $\mu_E = 0.02$, with $\tau = 0$. Here $\alpha \approx 0.00648216 < \mu_E$. Scalar interval endpoint and derivative tests enclose positive equilibria near $E_1 = 0.0261223656$, $N_1 = 99.8693882$, and $E_2 = 1.4058385649$, $N_2 = 92.9708072$. Interval Routh tests of the full four-state Jacobian give exactly one right-half-plane eigenvalue at the lower root and no right-half-plane eigenvalues at the upper root. Together with the stable zero-effort rest this provides genuine local bistability for this fully specified example. It does not prove a complete global basin separator. The analytic Jacobian agrees with independent high-precision differentiation to below 6×10^{-88}; full root boxes and Hurwitz minors are in `tether_equilibria_v42.json`.

These auxiliary parameters are illustrative mathematical choices, not field estimates. No universal ratio between $\mu_{\max}$ and α is asserted. The tether changes the equilibrium and can change Hopf/fold thresholds; the untethered thresholds do not transfer automatically.

S5. The Macro-Reduction Conjecture (hypothesis of Proposition 2)

This section registers the macro-reduction conjecture that underlies the main article's reduction from the five-state core to the working three-state system. The conjecture's slow-fast bound is stated, its five hypotheses are itemised, and the present status of each is given. Two hypotheses are established and one remains open; the remaining two are standard regularity hypotheses.

Conjecture S5.1 (Five-state macro reduction). The five-state core (stock, active pool, detritus, memory, effort) admits a slow-fast reduction whose slow subsystem tracks the working core with $\sup_{[0,T]} \|\mathbf{x}^\varepsilon - \mathbf{x}^0\| \le C(\varepsilon + \omega_A T)$ and slow-manifold tracking $\|\mathbf{y}^\varepsilon - \mathbf{h}(\mathbf{x}^\varepsilon)\| \le C(\varepsilon + \omega_A T) + Ce^{-\gamma_{\mathbf{y}} t/\varepsilon}$, under the following hypotheses:

- (H1) Index-1 DAE regularity.
- (H2) QSS manifold existence.
- (H3) Uniform normal hyperbolicity.
- (H4) Delay enters only in the slow variable.
- (H5) Lipschitz regularity.

Status of the hypotheses. The precise finite-time theorem — setting, hypotheses H1–H5, conclusion, and proof sketch (boundary layer on the fast clock with the delay as a slowly drifting parameter; retarded Gronwall closure) — underlies the status list below; two hypotheses are established and one remains open:

- **Fast-block uniform normal hyperbolicity — proved on a parameter set, quantified on the class.** At the QSS the fast Jacobian is block lower-triangular with eigenvalues $\{-\eta_{A,0} - \nu\tilde{\Delta},\ -(\eta_{T,0} + \eta_{T,1}\tilde{\Delta})\}$ and the 2×2 K–L block

  $$J_{KL} = \begin{pmatrix} -(1-\alpha)(\delta_K + \theta\tilde{\Delta}) & \beta\varsigma Q/L \\ \eta\alpha r_{L,0} L/K & -r_{L,0}(1-\eta\beta) \end{pmatrix},$$

  whose trace is $(\alpha - 1)(\delta_K + \theta\tilde{\Delta}) + r_{L,0}(\eta\beta - 1)$ and whose determinant is $r_{L,0}(\delta_K + \theta\tilde{\Delta})(1 - \alpha - \eta\beta)$ (the K-row uses the QSS identity $\varsigma Q/K = \delta_K + \theta\tilde{\Delta}$). Hence the fast block is Hurwitz, uniformly in $\tilde{\Delta} \ge 0$ and in the slow state, **if and only if** $\alpha + \eta\beta < 1$, and the QSS is then unique. Verified: sympy reproduces the formulas; central differences reproduce the analytic 4×4 Jacobian to 3.9×10^{-7} over 400 draws × QSS roots; an 11,792-draw sweep on the declared anchored class finds 1.9% of the class violating $\alpha + \eta\beta < 1$ — exactly the subset where the block is genuinely unstable (worst $\mathrm{Re}\,\lambda = +3.9 \times 10^{-3}$; multi-root QSS confined to the same subset) — and on the restricted classes $\{\alpha + \eta\beta \le 0.95\}$ / $\{\le 0.90\}$ the worst eigenvalue real part is -4.1×10^{-4} / -7.1×10^{-4} over 56,127 / 53,859 draws. The hypothesis therefore holds on the restricted class with the quantified margin, and the 1.9% violating subset of the declared class is excluded with the worst violation reported.
- **DAE regularity — theorem.** CES demand is strictly decreasing in price, so the algebraic equation is index-1 with a uniform bound (H1 holds).
- **Memory-kernel condition — open.** Infinite-time persistence (Fenichel-type for the delayed system) requires the spectral gap and compactness of the memory kernel of the infinite-dimensional semigroup, which is not verified; the reduction remains a finite-time statement, and the $O(\omega_A T)$ term (0.25–0.39 on one oscillation period) means the bound does not control the multi-century objects on one period.

The "exact" boundary. Exactness survives only for the strict-specialisation triangular projection (Theorem 2.1 of the main text's register: the core spectrum is a literal factor, proved). The macro reduction is leading-order with $O(\varepsilon + \omega_A T)$ corrections on finite intervals; the frozen-A step is the inner-approximation bound of Proposition 1 (trajectory-level, with $\varepsilon_A = A_0/A^* \approx 2.5 \times 10^{-3}$); and the recorded 3.2%/0.2% Hopf shifts are directly computed spectral sensitivities, 12.7 times the parameter ratio ε_A, not consequences of the trajectory bound. For computational methodology the upshot is that the working-core readings are leading-order correct on finite horizons, and multi-century extrapolation awaits closure of the memory-kernel condition.

S6. Open Problem Register

This register collects the six open problems whose resolution would close declared gaps in the main article. Each item is stated at conjecture status with its missing components identified; none is asserted as a theorem.

1. **Periodic-orbit-fold persistence under typed coupling.** A transverse fold of periodic orbits in a registered retarded subsystem persists under sufficiently small typed vector coupling and compatible additional fixed delays when the orbit has the required spectral gap and the infinite-dimensional Poincaré map is sufficiently smooth. Missing: a verified fold baseline, formulation of the coupled system on a common phase space, and verification of transversality, spectral separation, and regularity. Disproof route: arbitrarily small admissible coupling destroying the fold through resonance, boundary contact, loss of regularity, or a competing centre direction. (Note: transverse normal hyperbolicity does not imply normal hyperbolicity of the fold orbit itself.)
2. **RFDE/hybrid transition persistence.** Persistence of a reduced hybrid/RFDE nonlinear transition requires well-posedness of the semiflow, fast difference-operator contractivity, spectral separation, centre-manifold reduction, transverse Poincaré-map conditions, regular coupling, and preservation of material feasibility and safety. The fast difference-operator contractivity condition applies to particular neutral/difference formulations and is not a universal RFDE condition.
3. **n-patch super-equilibrium equivalence.** For n coupled patches with cooperative diffusion, the viability kernel is nonempty iff the minimal-harvest field has a super-equilibrium in the admissible set (two-patch instances verified); for type-(U) patches with product controls this is equivalent to the existence of an equilibrium. The equivalence holds for cooperative transport and fails in general for competitive or predator–prey couplings.
4. **Variable-time delayed-hybrid information kernel.** A nontrivial class of variable-event delayed hybrids with compact piecewise-history phase space, uniform break and jump budgets, resets, and bounded-error partial observations admits an exact compact information process and closed information-state tube kernel. Two independent gaps: the piecewise-history topology, and the closure of bounded-error observation fibres.
5. **Restricted delay-separation principle.** If ecological observation, assessment, decision, and deployment modules are independently identified and enter a locally linear loop as a product of stable transfer operators, their phase contributions can be separated from sufficiently rich input–output data up to declared structural symmetries. Missing: structural identifiability of factored transfer functions and persistence under feedback.
6. **Exergy-gated suppression.** For a declared class of autocatalytic extractive controllers, sufficiently low deployable exergy reduces the loop gain below every admissible Hopf-frequency modulus condition (a prospective member of the loop-gain exclusion family of Theorem 6 of the main article). Not universal: depletion of institutional capacity may also disable protective action or create hysteresis. Stated at conjecture status; no class, bound, or proof is claimed.

S7. Statement Inventory

A per-statement inventory (each statement of the main article with its status — theorem, conditional statement, numerical result, conjecture, or definition — and its evidence source: displayed proof, interval certificate, re-execution-verified computation, or numerical record) is available from the authors and accompanies the deposited material. The certification hierarchy of the main article is instantiated claim by claim: proved theorems (the cubic, the even-pairs algebra, the no-Hopf theorem, the monodromy formulas, the loop-gain exclusions); interval-certified local spectrum (the Hopf enclosures of S1); re-execution-verified nominal computations (continuation, multipliers, basins, Lyapunov coefficients); and declared-numerical or open items (the fold classification, the registry variants); the reduction conjecture's fast-block spectral condition is verified (S5), with the memory-kernel condition open.

S8. Registered Algebraic Template: The Delayed-Effort Audit Core

The audit core is the delayed-effort template in which deployed effort responds to a delayed decline signal through a deployable-capital gate $g_0 \in (0,1)$, with a linear loss $\mu_E E$ placed outside the multiplicative gate and a capital state K_0. Its algebraic consequences are registered here because the main article's effort laws are specialisations of this template.

Proposition S8.1 (Effort sensitivity coefficients). At a regular interior equilibrium of the template with deployable-capital gate $g_0 \in (0,1)$, the effort sensitivities are $C_Z = h_0 g_0 \eta E^*/\Delta_{\mathrm{ref}}$ and $C_K = \mu_E E^*(1-g_0)/(K_0 g_0)$.

Proof. The displayed coefficients are the declared partial derivatives of the registered template effort law with respect to the delayed signal and to deployable capital, evaluated at the regular interior rest; both are elementary differentiations of the template algebra, and the gate factor g_0 multiplies the signal term as displayed. □

C_Z is the local sensitivity of effort growth to the delayed decline signal, multiplied by the gate; C_K becomes a damping pathway only through its coupling to the separate capital dynamics, not by its sign alone. The factor $1/g_0$ cannot be extrapolated to $g_0 = 0$, where the assumed regular positive interior branch and the algebra behind the coefficients need not persist.

Proposition S8.2 (Interior effort upper bound). An interior equilibrium with $Z^* = 0$ and $E^* > 0$ must satisfy $h_0 g_0(\delta_0 - \eta(E^*)^2/E_{\max}) = \mu_E E^* > 0$, hence $E^* < \sqrt{\delta_0 E_{\max}/\eta} \approx 0.573$ at $\delta_0 = 0.3$, $E_{\max} = 1$, $\eta = 0.914$.

Proof. At the rest, $Z^* = 0$ kills the signal term, and the effort equation reads $h_0 g_0(\delta_0 - \eta(E^*)^2/E_{\max}) - \mu_E E^* = 0$; positivity of the right-hand side of the first equality gives the strict bound, and solving the bound at the declared values gives the displayed number. □

This is an equilibrium consequence of placing the linear loss outside the multiplicative gate, not a failure of boundary invariance ($\dot{E} \geq 0$ at $E = 0$ and $\dot{E} < 0$ at $E = E_{\max}$ both hold); moving the loss terms defines a different model and requires renewed analysis.

S9. The Maturation-Delayed Recruitment System: Registration Records

All records of Section 9 of the main paper are certified by the 21-gate registration campaign (`rerun_campaigns/campaign_p4_dr_registration.py`); the gate log is `rerun_campaigns/results/p4_dr_registration_gates.txt` and the fine-map table `rerun_campaigns/results/p4_dr_finemap_bands.csv`. Every gate passed.

S9.1 Sign-flip certificates (G1)

At the gated Candidate A coefficients ($A_N = -0.01791$, $A_E = -0.089552$, $B_N = 0.001791$, $B_E = 0.008955$, $C_E = -0.059518$, $C_Z = 1.785019$, $d = 0.2$), the registered fundamental pair is reproduced exactly (3.666149 / 150.358477). Flipping C_Z gives fundamental delays 128.374373 / 70.696578 (the paper's displayed 128.374 / 70.697), by the branch mechanism $3.666149 + \pi/\omega_1$ and $150.358477 - \pi/\omega_2$ with $(\omega_1, \omega_2) = (0.0251915, 0.0394366)$; the flipped linearisation has loop gain 1.01922; both shifted imaginary-axis crossings are simple and transverse ($\mathrm{d}\,\mathrm{Re}\,\lambda/\mathrm{d}\tau = -6.07 \times 10^{-6}$ and $+1.83 \times 10^{-5}$).

S9.2 Fine-map institutional bands (G3)

At $\eta = 0.914$ the $\tau = 0$-stable two-crossing bands are reproduced with refined edges, within the recorded fine-map grid resolution of ± 0.03:

gate g	recorded band	re-run band	points in band
1	$(1.565, 1.585)$	$(1.5643, 1.5857)$	10
2	$(0.77, 0.81)$	$(0.7669, 0.8049)$	15
3	$(0.50, 0.55)$	$(0.4919, 0.5466)$	20
5	$(0.28, 0.33)$	$(0.2603, 0.3324)$	26

All eight recorded band cells (four band centres and four edges) are stable on the wide characteristic mesh with two delay crossings present (rightmost root ≈ -0.06).

S9.3 Mesh-range caveat (G3-probe)

The fine map's characteristic scan is a bounded mesh ($\sigma \in [-0.15, 0.08]$); the probe documents a mesh-range blind spot. At ($g = 5$, $\eta = 3.0$, $r \approx 1.54$–1.60) the narrow mesh reports an apparent $\tau = 0$-stable crossing cell; the wide mesh locates the true rightmost root at $+0.2445$ (unstable), and the nonlinear trajectory diverges — the cell is a cohort regime of the delayed plant, not a $\tau = 0$-stable band. The recorded table's absence of institutional $\tau = 0$-stable cells at $\eta = 3.0$ is thereby confirmed; the apparent band is withdrawn as a classification artefact, and band cells in the paper are claimed stable only where the wide mesh confirms them.

S9.4 Nonlinear ground truth (G4)

configuration	recorded	verbatim re-run
($r = 0.02$, $g = 5$, $\tau = 0$) cohort cycle	$P = 358.8$, amplitude 66.9	$P = 358.7$, amplitude 66.9
($r = 0.5$, $g = 5$, $\tau = 0$) cohort cycle	$P \approx 20$	$P = 20.1$
($r = 0.3$, $g = 5$, $\tau = 0$)	stable, two crossings	stable, two crossings
($r = 0.3$, $g = 5$, $\tau = 10$) institutional cycle	$P = 16.96$, amplitude 8.66	$P = 16.95$, amplitude 9.36 (tail-window dependent)
($r = 0.3$, $g = 5$, $\tau = 21$)	stable	stable (amplitude 0)
($r = 1.57$, $g = 1$, $\tau = 2.5$, $\eta = 3$) band centre	$P = 4.0$, amplitude 3.5	$P = 4.00$, amplitude 3.49
($r = 0.8$, $g = 2$, $\tau = 5.5$, $\eta = 3$) band centre	$P = 8.04$, amplitude 12.4	$P = 8.05$, amplitude 12.35

S9.5 Compute-core self-check (G5)

The compute core's gated Candidate A Hopf pair reproduces the registered P4 certificates 3.666149 / 150.358477 (period rows 249.4 / 159.3 yr), closing the loop between the archived stage machinery and the paper's own delay certificates.

S10. Expanded Proofs: Corollary 5.1 and Proposition 6.1

This section records the two elementary facts on which the appended proof of Corollary 5.1 rests, and the branch-safe phase evaluation used in the appended proof of Proposition 6.1. Both are standard; they are stated here for completeness so that the main article's proofs are self-contained.

S10.1 The certificate map of Corollary 5.1

The proof appended to Corollary 5.1 rests on two elementary facts, stated here for completeness. (a) *Uniform tail bound.* With A_N, d, C_E continuous in $\chi_m \in [0,1]$ (hypothesis (i)) and $|C_m| + |C_p| \le |C_Z^{\mathrm{mob}}| + |C_Z^{\mathrm{prot}}|$, there is $\overline{\omega}$ such that $(|C_m| + |C_p|)|L(i\omega)|/|(i\omega - A_N)(i\omega + d)(i\omega - C_E)| < F(0)/2$ for all $\omega > \overline{\omega}$ and all $\chi_m \in [0,1]$ — the ratio is $O(\omega^{-2})$ with O-constant uniform on the compact parameter interval. (b) *Continuity of the maximum.* A maximum over a fixed compact set of a jointly continuous function is continuous in the parameter. Both are standard; (a) is where hypothesis (i)'s compactness is used, and (ii) is where the denominator's nonvanishing is used.

S10.2 The phase formula of Proposition 6.1

The fundamental-delay formula used in the appended proof of Proposition 6.1 is the branch-safe phase evaluation: for the characteristic function $\Delta(\lambda, \tau) = P(\lambda) - C_Z L(\lambda) e^{-\lambda\tau}$, the delay families are

$$\tau^{(k)}(\omega) = \frac{\varphi(\omega) + 2\pi k}{\omega}, \qquad \varphi(\omega) = -\arg \frac{P(i\omega)}{C_Z L(i\omega)},$$

with the fundamental branch the smallest positive representative of each family. Flipping C_Z adds π to the phase numerator, translating each family by the odd half-period π/ω: the lower family's fundamental branch moves from 3.666149 to $3.666149 + \pi/\omega_1$, the upper family's from 150.358477 to $150.358477 - \pi/\omega_2$ (the branch that keeps each family fundamental). Simplicity and transversality at the shifted points are certified numerically ($\partial_\lambda \Delta \neq 0$; $\mathrm{d}\,\mathrm{Re}\,\lambda/\mathrm{d}\tau = -6.07 \times 10^{-6}$ and $+1.83 \times 10^{-5}$), which establishes a linear spectral diagnostic, not a nonlinear constitutive law for the reversed-gain institution. —

S11. MPF Material

Material from the main article's Section 8.3 (the MPF paragraph) and Section 10.4. Every value below is reproduced verbatim from the main text; nothing is recomputed, and the main text retains one-sentence pointers carrying the key numbers ($\eta_{\mathrm{crit}} \approx 2.337$; inter-excursion-interval coefficient of variation 1.58 and return-map anticorrelation $r = -0.47$; the more-than-300-parameterisation screen).

S11.1 The η_{crit} sweep and the pair-birth structure

The absence of a baseline Hopf is parametric: Hopf roots first appear at $\eta_{\mathrm{crit}} \approx 2.337$, with two interleaving pairs over $\eta \in (2.337, 3]$ (at $\eta = 2.5$: ≈ 0.6, 54.2, 92.9, 113.1 yr; at $\eta = 3.0$ one pair spans ≈ 4.5–41.2 yr; at the out-of-range $\eta = 10$: 17.568/18.362 yr with a supercritical-consistent onset, exponent 0.59, inferred). The pair-birth structure behind the interleaving is registered: the large-delay pair is born at $\eta_{\mathrm{crit}} \approx 2.337$ ($\tau_- \approx 71.2$, $\tau_+ \approx 72.9$ yr) and migrates downward as η rises, while the small-delay pair is born at $\eta \approx 2.454$ with $\tau_- \to 0$ at its onset.

S11.2 The slow-fast intermittency diagnostics ($\eta = 10$, above τ_+)

Above τ_+ at $\eta = 10$ the attractor is diagnosed as irregular slow-fast intermittency, with diagnostics inconsistent with a simple period-one or low-period orbit — the underlying global bifurcation mechanism (a homoclinic connection among the candidates) is unresolved. The diagnostics: a broad onset-interval spectrum with no sharp peak, a thin map-like Poincaré section on Z-crossings, inter-excursion-interval coefficient of variation 1.58, and return-map anticorrelation $r = -0.47$ — with the large-amplitude time fraction rising monotonically from 0% at $\tau = 18.4$ to 100% by $\tau \approx 22$ and no sharp second threshold separating quiet and captured regimes; the present diagnostics neither identify a homoclinic mechanism nor exclude torus breakdown.

S11.3 The sigmoid-gated effort screen (more than 300 parameterisations)

A sigmoid-gated effort variant of the same ecological core — the saturating gate $\sigma(Z_\tau/Z_0)$ of the main text's Section 10.4, whose linearisation at equilibrium decreases in deployed effort and contains no factor $\eta E^*/\Delta_{\mathrm{ref}}$ — was screened across more than 300 parameterisations (Newton eigenvalue tracking, joint modulus minimisation, nonlinear integration) without finding a genuine delay-induced Hopf — a numerical negative result over the sampled domain, not a structural impossibility theorem. The screen's loop-gain reading (it does not establish that autocatalysis is necessary for a Hopf pair, and the gated law (1) is not of this class) is stated in the main text's Section 10.4; this record is deposited here.

S12. Statement Cross-References

The main article and this supplement use the same substantive statements: the cubic modulus theorem is main Theorem 4.1; weighted delay exclusion is Proposition 5.2 and Corollary 5.1; the iso-gain spectral diagnostic is Proposition 6.1; the reviewed-map identities are Theorem 7.1 and Proposition 7.1. Classical nonlinear Hopf verification is Theorem 5.1/S13; cadence-independent protective stability, the contractive review class and nonlinear stock-floor protection are Theorems 7.2–7.4/S14. The discrete folds and continuum requirements are S3, and numerical branch/capture evidence is S15.

S13. Verified nonlinear Hopf coefficients

S13.1 Derivatives of the declared retarded equation

Write $D(N, E) = qEN - rN(1 - N/K)$ and $\Phi_\kappa(D) = \log(1 + e^{\kappa D})/\kappa$. At an interior steady state $D = 0$, $Z = \delta$, so $\Phi' = 1/2$, $\Phi'' = \kappa/4$, and $\Phi''' = 0$. Use extended directions $u = (u_N, u_Z, u_E, u_\zeta)$, where ζ is the

delayed memory argument. Then

$$D_1(u) = (qE^* - r + 2rN^*/K)u_N + qN^*u_E,$$

$$D_2(u,v) = 2ru_Nv_N/K + q(u_Nv_E + u_Ev_N).$$

For $f = (f_1, f_2, f_3)$, define $B = D^2 f$ and $C = D^3 f$, without factorials. Their ecological components are

$$B_1 = -D_2, \quad C_1 = 0, \quad B_2(u,v) = \frac{D_2(u,v)/2 + \kappa D_1(u)D_1(v)/4}{\tau_m},$$

$$C_2(u,v,w) = \frac{\kappa}{4\tau_m}\big[D_2(u,v)D_1(w) + D_2(u,w)D_1(v) + D_2(v,w)D_1(u)\big].$$

At the registered unit references, put $M = E_{\max}$ and $g = 1 - E^*/M$. The nonzero gated effort derivatives are

$$f_{EE} = -2\eta(1+\delta)/M + 6\eta E^*/M^2, \quad f_{E\zeta} = \eta - 2\eta E^*/M - \delta_0/[M(1+\delta)^2],$$
$$f_{\zeta\zeta} = -2\delta_0 g/(1+\delta)^3, \quad f_{EEE} = 6\eta/M^2, \quad f_{EE\zeta} = -2\eta/M,$$
$$f_{E\zeta\zeta} = 2\delta_0/[M(1+\delta)^3], \qquad f_{\zeta\zeta\zeta} = 6\delta_0 g/(1+\delta)^4.$$

These entries and symmetry determine B_3, C_3. Without the outer gate, $f_{EE} = -2\eta/M$, $f_{E\zeta} = \eta$, $f_{\zeta\zeta} = -2\delta_0/(1+\delta)^3$ and $f_{\zeta\zeta\zeta} = 6\delta_0/(1+\delta)^4$; the other effort third derivatives vanish.

S13.2 RFDE homological formula and convention

Let A_0, A_1 be the instantaneous and delayed Jacobians and $\mathcal{D}(s) = sI - A_0 - A_1e^{-s\tau}$. At $\lambda = i\omega$, choose $\|v\|_2 = 1$ and the complex left row p so that $p\mathcal{D}(\lambda) = 0$ and $p\mathcal{D}'(\lambda)v = 1$. A mode h at frequency s is lifted to $L_sh = (h_N, h_Z, h_E, e^{-s\tau}h_Z)$. Set $V = L_\lambda v$ and

$$h_{20} = \mathcal{D}(2\lambda)^{-1}B(V,V), \qquad h_{11} = \mathcal{D}(0)^{-1}B(V,\overline{V}),$$

$$G_{21} = p\big[C(V,V,\overline{V}) + B(\overline{V}, L_{2\lambda}h_{20}) + 2B(V, L_0h_{11})\big].$$

The physical-time coefficient is $\ell_1 = \operatorname{Re} G_{21}/2$. In a unit-frequency time convention it is divided by ω. This positive conversion does not change criticality. Only the right vector has unit Hermitian norm; normalizing both vectors independently would generally violate the required adjoint pairing.

For gated A, the physical-time coefficients are $5.74790328 \times 10^{-5}$ and $3.54919608 \times 10^{-4}$, both interval-positive. The real eigenvalue crossing derivatives have opposite strict signs, approximately $-9.48314396 \times 10^{-5}$ and $5.50936083 \times 10^{-6}$. Interval bounds exclude zero from $|\partial_\lambda\Delta|^2$, $\Delta(0)$ and the second-harmonic resolvent. The cubic has precisely two positive roots; the remaining root is negative. The first recurrence of the lower-frequency family lies beyond 253 yr and hence beyond the upper fundamental crossing. There is therefore no additional imaginary family at either point. These facts establish the classical smooth-retarded-equation Hopf hypotheses and the two subcritical local branches. They do not construct a finite branch-radius enclosure.

For ungated B, the corresponding coefficients are approximately -9.84424×10^{-5} and 2.18520×10^{-3}. The same interval simplicity, transversality, resolvent and recurrence tests verify their negative/positive signs, yielding the supercritical lower and subcritical upper crossing. The all-variant scalar table below does not assert nonlinear cubic certification for the other two gating combinations.

The scripts `hopf_cubic_v42.py` and `all_scalar_hopf_v42.py` use `mpmath.iv` at 70 and 80 decimal digits with exact decimal parameter strings and outward-serialized endpoints. High-precision automatic directional differentiation of the original nonlinear formulas agrees with the analytic B, C derivatives to less than 3.5×10^{-84}. Full intervals are in `hopf_cubic_interval_v42.json` and `hopf_cubic_B_interval_v42.json`. No radii-polynomial Hopf neighbourhood or continuum fold lift is asserted.

Candidate	Gate	Fundamental delay enclosure (years; displayed outer rounding)
A	gated	[3.666149014, 3.666149015]
A	gated	[150.358477310, 150.358477311]

Candidate	Gate	Fundamental delay enclosure (years; displayed outer rounding)
A	ungated	[6.881410630, 6.881410631]
A	ungated	[132.374903454, 132.374903455]
B	gated	[5.512840731, 5.512840732]
B	gated	[80.424526714, 80.424526715]
B	ungated	[6.213598734, 6.213598735]
B	ungated	[76.290635687, 76.290635688]

S14. Protective review margins and nonlinear viability

For $a = -A_N > 0$, the ecological effort-to-memory impulse response is $g(t) = B_E(de^{-dt} - ae^{-at})/(d-a)$. Its single sign change occurs at $t_0 = \log(d/a)/(d-a)$. Integrating its positive and negative parts gives

$$\|g\|_1 = \frac{2B_E}{d}(a/d)^{a/(d-a)} \le 2B_E/d = qN^*.$$

At $a = d$ the continuous limit is $2B_E/(ae)$. The protective effort-filter gain is $k/c = E^*/(Z_{\mathrm{ref}} + \delta)$, independent of $\eta_p > 0$. Their product is enclosed near $0.1380266770 < 1$ at Candidate A. Weighted continuous/discrete small gain proves the exact/native all-cadences statements of Theorem 7.2. Each cadence and latency is fixed and finite; the theorem does not assert a uniform exponential rate as they tend to infinity.

The θ-review recursion has gain k/c when its scalar multiplier is nonnegative. When that multiplier is negative, its gain is $kT_r/[2 + (2\theta - 1)cT_r]$, bounded by $k/[c(2\theta - 1)]$. Thus $\theta > 0.569014$ suffices at the point vector; $\theta \ge 0.57$ is a conservative decimal choice. Explicit Euler is outside this unconditional range. Across the main-text mathematical parameter box, the conservative gain bound is less than 0.274, so exact/native review and $\theta \ge 0.64$ review are certified. The 0.57 value is not claimed uniformly over that whole box.

For nonlinear safety, the quota bound $C(Z) \le E_0$ and positive effort-relaxation factor make $E \le E_0$ invariant when the initial effort lies below it. Scalar logistic comparison gives, more generally,

$$N(t) \ge \frac{N_b N(0)}{N(0) + [N_b - N(0)]e^{-(r - q\overline{E})t}}, \qquad N_b = K(1 - q\overline{E}/r) > 0,$$

for $N(0) > 0$, whenever effort remains below $\overline{E}$. In particular $\min\{N(0), N_b\}$ is a permanent stock lower bound, and $\liminf N(t) \ge N_b$. This concerns stock retention/recovery; it need not imply convergence of the full state to one equilibrium. For the calibrated hyperbolic cap $E_0 = 2.2344650544$ and $N_b = 88.8276747280$. The main-text box, with the additional restriction $E_{\max} \in [26, 34]$, has a common conservative floor 75.74. These are dimensioned model-state bounds, not estimated fishery reference points or confidence intervals.

The nonlinear exact quota update in Theorem 7.4 maps an effort value to the interval between itself and the sampled cap. Its equilibrium Jacobian has entries $e^{C_E T_r}$ and $C_Z(e^{C_E T_r} - 1)/C_E$, exactly the review factor of Proposition 7.1. Thus the stock-floor result and the linear exact-review theorem refer to a compatible nonlinear realization, not unrelated maps. The native hold preserves the same effort interval continuously, including between reviews and under delayed nonnegative observations.

For the eventual-stability theorem, the frozen ecological flow tends to $u\mathbf{e}_3^\top$, with $u = (-A_E/A_N, 0, 1)^\top$. A bounded measurement gain and a limiting effort multiplier of modulus less than one yield the limiting spectrum $\{0, 0, \alpha_\infty\}$. Exact and $\theta > 1/2$ update rules satisfy this condition; consistency alone does not. At the registered native parameters A_Z is Hurwitz, and its limiting rank-one operator is nilpotent by the zero DC gain. These are spectral conclusions; the additional nonlinear protective theorem is what supplies a declared stock-floor operating region.

S15. Periodic-branch continuation and nonlinear capture

S15.1 Interior-coordinate equations and collocation

Set $u = \log N$, $v = \log[E/(M - E)]$, $E = M/(1 + e^{-v})$ and retain Z. Then

$$\dot{u} = r(1 - e^u/K) - qE,$$

$$\dot{Z} = [\Phi_\kappa(e^u(qE - r + re^u/K)) - Z]/\tau_m,$$
$$\dot{v} = \eta(Z_\tau - E/M) + \delta_0 Z_\tau/[E(1 + Z_\tau)].$$

The last equation is the original gated effort equation divided by $E(1 - E/M)$; it contains no regularizing floor. The transformations enforce $N > 0$ and $0 < E < M$ analytically. The memory coordinate is not clipped.

Odd-grid Fourier collocation uses $DX - Tf(X, S_{\tau/T}Z) = 0$ and the phase condition $\sum_j \sin(2\pi j/n)u_j/n = 0$. Odd node counts avoid an ambiguous even-grid Nyquist mode. The inherited physical seed at $\tau = 64.4382853855$ is corrected at 129 nodes; it is not assumed to solve the transformed equations exactly. Natural-parameter predictor/corrector continuation raises resolution to 257 and 513 nodes when off-grid defects require it. The archived output contains 131 records ending at 155 yr. A coordinate chart and a small residual do not constitute a validated continuum enclosure.

S15.2 Mesh and spectral evidence

Delay (yr)	Period (yr), 1025 nodes	Dominant nontrivial modulus
64.438285	72.849447806	0.943549239
80.000000	87.986215181	0.929942048
100.000000	108.012663376	0.910734381
120.000000	127.793114270	0.870034568
149.500000	156.828957843	0.800666663
155.000000	162.239392603	0.786978704

Each selected point is corrected successively at 257, 513 and 1025 nodes and checked on 4097 physical off-grid samples. Period changes from 513 to 1025 nodes are below 10^{-9} yr; componentwise physical defects are below 1.3×10^{-11}. The table gives the 1025-node period and the dominant nontrivial Floquet modulus at step 0.01 yr. Floquet calculations are repeated at 0.04, 0.02 and 0.01 yr. A method-of-steps variational map with Arnoldi iteration retains the phase multiplier and identifies it using the orbit tangent. These floating spectral calculations do not enclose the complete infinite-dimensional spectrum. No rigorous global fold or uniqueness conclusion follows.

S15.3 Fresh nonlinear waveform comparisons

H1 denotes the constant history $(N, Z, E) = (90, \delta, E_A^*/2)$ and H2 denotes $(5, \delta, 15)$. They are distinct from the response-sign taxonomy of S4. For each selected history/delay, the nonlinear method-of-steps simulation uses the transformed equations, fixed steps 0.04 and 0.02 yr, and no clipping. The period and all three amplitudes of the reference orbit remain fixed; only one common phase shift is fitted. RMS errors are normalized by the three reference component ranges.

History	Delay	Horizon	Step	Normalized RMS	Waveform match
H1	149.5	100000	0.04	1.66983e-07	yes
H1	149.5	100000	0.02	8.29171e-08	yes
H2	149.5	40000	0.04	0.551139	no
H2	149.5	40000	0.02	0.551139	no
H2	155.0	40000	0.04	3.28874e-07	yes
H2	155.0	40000	0.02	8.21467e-08	yes

H1 at 149.5 matches after 100000 yr, while H2 at the same delay does not match within 40000 yr. H2 at 155 matches after 40000 yr. The finer-step normalized RMS of each matching case is below 8.4×10^{-8}; the nonmatching case is about 0.5511. The negative result is part of the evidence. The comparison establishes a numerical connection and two observed captures, not generic capture, an exact infinite-horizon basin boundary, or exhaustive attraction basins.

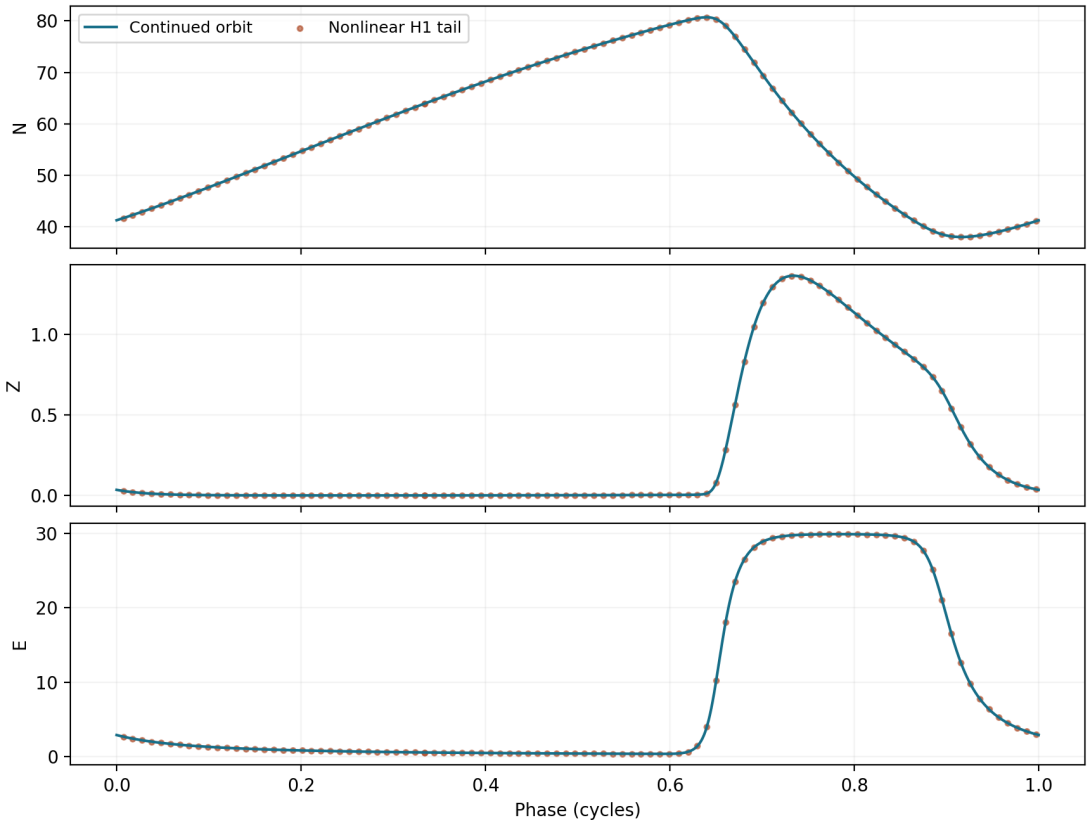


Figure 1: Figure S15. Continued orbit and independently integrated H1 tail at delay 149.5 years. The overlay uses one phase shift, with fixed period and amplitudes.

S16. Within-stock institutional-event evidence

S16.1 Extraction and endpoint protocol

The observational unit is a documented advice/action episode within North Sea autumn-spawning herring, not a national aggregate. The targeted source pair is Simmonds (2007), Table 1, pp. 688–689, and the Cefas account by J.H. Nichols (1999), *Saving North Sea herring*. The first reconstructs historical ICES/NEAFC records; the second supplies agency chronology for 1995–1996. The source pair was selected to test temporal plausibility within the small-pelagic class. This is a transparent retrospective extraction, not a preregistered or representative sample.

Events are stored with separate advice, restrictive implementation, closure and reopening labels in `institutional_events_v42.csv`. A year has January 1–December 31 bounds; a month retains its first/last dates. For a directed advice-to-action clock, interval subtraction propagates precision and a causal lower bound of zero. The 1996 action is constrained to follow the urgent May advice. This also tightens the action lower bound for the October 1995 advice clock. Day differences are divided by 365.25; displayed bounds are rounded outwards. Calendar bounds are not confidence intervals. Multiple clocks sharing an action or episode are dependent.

The 1965 general recommendation, 1970 specific TAC recommendation, 1976 first restrictive TAC and 1977 directed closure are extracted from the separately typeset rows, not inferred from a figure. Nominal earlier TAC agreements are not relabeled as restrictive implementation. The October 1995 and May 1996 advice events and implementation during 1996 are separately recorded. No exact implementation day is assigned from the date of a negotiation meeting. The general closure ending in 1983 remains a distinct endpoint; it is not pooled with reports of earlier partial reopening or used as an advice-to-action delay.

The derived clocks are approximately $[10, 12]$, $[5, 7]$, $[6, 8]$, $[0.50, 1.26]$ and $[0, 0.67]$ yr with the definitions in the main text. They constrain which historical clock is compatible with a predicted time window. Five clocks from two episodes do not supply an empirical governance-lag distribution or identify the model's constant deployment delay. Post hoc overlap with the $g = 2$ window is a compatibility check, not a fitted

likelihood or prediction score.

S16.2 Identification and the next dynamic test

A dynamic fishery test requires a stock observation operator (for example, spawning biomass versus total renewable stock), a unit-consistent effective-effort series, a separately measured institutional response/process signal, recruitment and major environmental covariates, and a specified response-law interpretation. The transformation $E \mapsto sE$, $q \mapsto q/s$, $E_{\max} \mapsto sE_{\max}$, $\delta_0 \mapsto s\delta_0$ and $E_0 \mapsto sE_0$ leaves N, catch and the scaled model dynamics invariant. Stock/catch data alone therefore cannot identify both effort units and catchability.

The different TAC quantities in retrospective accounts are not pooled without scope and assessment-vintage reconciliation. Spawning-stock observations are not silently used as the model total stock. No plotted series is digitized, no stock values are generated from projections, and no event lag is entered as an observed DDE parameter. The present evidence supports temporal discrimination within one stock, not dynamic validation or global calibration. A prospective dynamic test should fix its observation mapping and competing recruitment/environmental model before estimation, separate training and evaluation periods, and propagate process/observation uncertainty. S20 supplies quantitative assessment-based dynamic tests; its observation gauge, non-confirmatory amendment, ecological competitors and negative recursive endpoints are separate from these event-clock data.

S16.3 Sources

Simmonds, E.J. (2007). Comparison of two periods of North Sea herring stock management: success, failure, and monetary value. *ICES Journal of Marine Science* **64**, 686–692. doi:10.1093/icesjms/fsm045. Publicly accessible source: https://www.vliz.be/imisdocs/publications/222472.pdf .

Nichols, J.H. (1999). *Saving North Sea herring.* Cefas, Lowestoft, 4 pp. Agency publication record and account: https://www.cefas.co.uk/data-and-publications/publications/list/publication-abstract/?id=12671 .

Hocherman, L., Trop, T., Ghermandi, A. (2025). Time lags in environmental governance: A critical review. *Ambio* **54**, 2042–2059. doi:10.1007/s13280-025-02211-y. This heterogeneous synthesis motivates endpoint care; it is not used as a 2–13-year empirical delay distribution.

S17. Complete equilibrium root counts for every positive delay

At a mobilising registered vector, write

$$\Delta(\lambda,\tau) = P(\lambda) - k\lambda e^{-\lambda\tau}, \quad P = (\lambda+a)(\lambda+d)(\lambda+c), \quad a,c,d,k>0.$$

The exact coefficients come from the positive quadratic effort equilibrium, not rounded Jacobian values. The outward scalar calculation of S13 supplies two disjoint simple positive roots of

$$H(x) = (x+a^2)(x+d^2)(x+c^2) - k^2 x.$$

The cubic has two coefficient sign changes; the two verified roots exhaust its possible positive roots by Descartes' rule. Also $\Delta(0,\tau) = adc > 0$.

Proof of Theorem 5.2. At $\tau = 0$, put $p_2 = a+d+c$, $p_1 = ad+ac+dc-k$, $p_0 = adc$. The interval tests $p_2 > 0$, $p_0 > 0$, $p_2p_1 - p_0 < 0$ give the Routh first-column signs $+,+,-,+$, hence exactly two right-half-plane roots. For $\operatorname{Re}\lambda \geq 0$, each $|\lambda+s| \geq |\lambda|$ for $s>0$. At a root, $|\lambda|^3 \leq k|\lambda|$, uniformly in nonnegative delay. A half-disk of radius greater than $\sqrt{k}$ therefore encloses all unstable roots and has no roots on its curved boundary. Analytic root counting on each compact delay interval changes only at the two imaginary-frequency families; no unstable root can arrive from infinity, including as $\tau \downarrow 0$.

At an imaginary root, implicit differentiation gives

$$\left(\frac{d\lambda}{d\tau}\right)^{-1} = -\frac{P'(\lambda)/P(\lambda) - 1/\lambda + \tau}{\lambda}.$$

Taking real parts at $\lambda = i\omega$ yields

$$\operatorname{Re}\left[\left(\frac{d\lambda}{d\tau}\right)^{-1}\right] = \frac{1}{\omega^2+a^2} + \frac{1}{\omega^2+d^2} + \frac{1}{\omega^2+c^2} - \frac{1}{\omega^2} = \frac{H'(\omega^2)}{|P(i\omega)|^2}.$$

The nonzero interval signs prove simplicity and a direction independent of recurrence: the lower family removes two unstable roots; the upper adds two. Thus, off all critical delays,

$$U(\tau) = 2 - 2n_-(\tau) + 2n_+(\tau), \quad n_j = \#\{m \geq 0 : \tau_j + 2\pi m/\omega_j < \tau\}.$$

For $\tau > \max(\tau_-, \tau_+)$ not critical, $n_+ > (\tau - \tau_+)/P_+$ and $n_- < (\tau - \tau_-)/P_- + 1$. Hence

$$U(\tau) > 2\{\tau(1/P_+ - 1/P_-) - \tau_+/P_+ + \tau_-/P_-\}.$$

Since $P_+ < P_-$, this is positive above the cutoff in Theorem 5.2. The cutoff exceeds both fundamental delays at every registered vector. The finite interval event list resolves the order below it without overlapping event boxes. All stable intervals are therefore identified; critical endpoints are excluded. At simultaneous events beyond the cutoff the two simple pairs can be counted together, without creating an asymptotically stable critical point. Finally,

$$U(\tau) = \frac{\omega_+ - \omega_-}{\pi}\tau + O(1).$$

Candidate	Gate	All stable intervals (approximate endpoints, yr)	Eventual-instability cutoff (approx., yr)
A	gated	(3.666149014, 150.358477310); (253.082596558, 309.682275055)	409.775632
A	ungated	(6.881410630, 132.374903454); (269.252385064, 274.629634339)	280.997958
B	gated	(5.512840731, 80.424526714)	140.692703
B	ungated	(6.213598734, 76.290635688)	126.980515

The exact endpoint intervals, Routh inequalities, disk radii, cutoff bounds and complete finite event order are in `global_equilibrium_spectrum_v43.json`. This is an equilibrium spectrum result on an unbounded delay axis. It neither supplies the complete Floquet spectrum of a periodic orbit nor rules out coexisting nonlinear attractors at a stable equilibrium.

S18. Complete protective basin: invariance, comparison and nonlinear gains

Consider

$$\dot{N} = rN(1 - N/K) - qEN, \quad \dot{Z} = \{\Phi(-\dot{N}) - Z\}/\tau_m,$$

with $\Phi \geq 0$, $|\Phi(x) - \Phi(y)| \leq |x - y|$ and $\Phi(0) = \delta$. Let

$$C(Z) = E_0 Z_{\mathrm{ref}}/(Z_{\mathrm{ref}} + Z), \quad b(E) = \eta_p(1 - E/M).$$

The continuous/native effort law is $\dot{E} = b(E)(C(Z_{\mathrm{meas}}) - E)$. Native review reads $Z(t_k - \tau)$ and holds that signal until t_{k+1}, with fixed $T_r = t_{k+1} - t_k > 0$. For exact nonlinear frozen-effort review, the ecological/memory state flows with $E = E_k$ during the interval, and the command is

$$E_{k+1} = \Psi_{T_r}(E_k, C(Z_{\mathrm{meas,k}})),$$

where Ψ is the exact scalar nonlinear relaxation flow, not the flow of a linearisation. Start- or end-review timing only changes the bounded measurement age; the gain proof covers either stated causal convention. Delay is fixed and finite. A bounded initial memory history is nonnegative and need not be in equilibrium.

Invariance and persistence. At $N = K$ the stock derivative is nonpositive; at $N = 0$ it vanishes. Positive stock stays positive by the exponential variation formula. Memory is nonnegative and bounded, since $-\dot{N}$

is bounded on the physical stock/effort rectangle and Φ is globally Lipschitz. Since $0 \le C \le E_0 < M$, the interval $0 \le E \le \bar{E} = \max(E(0), E_0) < M$ is invariant. Consequently

$$b(E) \ge b_{\min} = \eta_p(1 - \bar{E}/M) > 0.$$

Continuous/native comparison gives $\limsup E \le E_0$. The exact scalar flow satisfies

$$\Psi_T(E, C) - C = (E - C) \exp\left(- \int_0^T b(E(s))\, ds \right),$$

so it is a convex relaxation with $0 < \alpha \le e^{-b_{\min} T} < 1$. This proves the same upper bound and limiting effort bound for reviewed commands and their physical holds. For every sufficiently small $\varepsilon > 0$, eventually $E \le E_0 + \varepsilon$. Logistic comparison then gives

$$\liminf N \ge K(1 - q(E_0 + \varepsilon)/r).$$

Letting $\varepsilon \downarrow 0$ proves $\liminf N \ge N_b > 0$.

Uniqueness. At a positive equilibrium, $\dot{N} = 0$, so $Z = \delta$. The effort equilibrium is $E = C(\delta)$ and the stock equilibrium follows. A fixed exact-review command must also equal its frozen quota because $\alpha < 1$. No other positive rest exists.

Asymptotic gains. Define $n = \log(N/N^*)$, $e = E - E^*$, $z = Z - \delta$ and the positive logarithmic-mean coefficient

$$a(t) = \frac{r}{K} \frac{N - N^*}{\log(N/N^*)}, \qquad a(t) = rN^*/K \text{ at } N = N^*.$$

The logarithmic mean lies between its positive arguments. Thus asymptotically $a \ge rN_b/K - o(1)$ and always $a \le r$. Also

$$\dot{n} = -a(t)n - qe, \qquad D = -\dot{N} = N\{a(t)n + qe\}.$$

All errors are ultimately bounded. Stable scalar convolution and the definition $L_x = \limsup_{t\to\infty} |x(t)|$ give

$$L_n \le \frac{qK}{rN_b} L_e, \qquad L_D \le K(rL_n + qL_e), \qquad L_z \le L_D.$$

The last step uses $|\Phi(D) - \Phi(0)| \le |D|$ and the memory filter's unit asymptotic gain. Since $|C(Z) - C(\delta)| \le L_C|Z - \delta|$, continuous/native tracking has $L_e \le L_C L_z$: outside any slightly enlarged quota-error interval, the tracking error points inward with rate at least $b_{\min}$. Every fixed bounded signal age preserves this limiting error bound after its initial history has washed out. For exact review, the same conclusion follows by iterating the convex scalar relaxation with uniform contraction factor at most $e^{-b_{\min} T_r}$.

Substitution yields

$$L_e \le qKL_C(1 + K/N_b)L_e = \Gamma L_e.$$

The strict gain forces $L_e = L_n = L_z = 0$.

Stability, not only attraction. Choose a neighbourhood of the equilibrium with $N > N_b - \varepsilon > 0$, and ε small enough that $qKL_C\{1 + K/(N_b - \varepsilon)\} < 1$. On any finite interval before a first exit, the same scalar estimates hold with suprema instead of limsups and additional terms bounded by the initial stock/effort errors and memory-history norm. Closing the strict gain bounds all three suprema by a fixed multiple of those initial errors. Sufficiently small initial errors therefore preclude the first exit. This proves Lyapunov stability. Together with the global attraction just proved, it establishes global asymptotic stability relative to the stated basin, without assuming differentiability of a general Lipschitz Φ.

Complete basin and limits. The faces $N = 0$ and $E = M$ are invariant for all three protected implementations and cannot converge to the positive equilibrium. Every other physical state with bounded nonnegative memory history is covered. Hence the basin is exact, not just an invariant subset or a finite set of histories. No nonconstant recurrent omega-limit set exists in it. The result supplies no uniform convergence rate near those excluded faces or as delay/cadence becomes arbitrarily large. Mixed/mobilising laws, Euler commands and unbounded signal ages do not inherit the proof. The inequality is sufficient, not necessary: $\Gamma \ge 1$ does not itself imply failure.

Outward constants. At the common protective point, $E_0 = E^*(1 + \delta)$, $Z_{\text{ref}} = 1$, $E_0 \approx 2.2344650544$, $N_b \approx 88.827674728$ and $\Gamma \approx 0.474997056688$. For the box stated in the main article, interval evaluation gives $\Gamma \le 0.686293596959 < 1$, with positive $1 - qE_0/r$ and $E_0 < M$ throughout. The source

is `global_protective_basin_v43.py`; `global_protective_basin_v43.json` stores exact outward endpoints. The box treats E^* and δ as protective-controller design coordinates and sets $E_0 = E^*(1+\delta)$; it is not a fitted ecological/institutional confidence region.

S19. Infinite-Fourier residual validation for approximate folds

Let $s = t/T$ and $\sigma = \tau/T$. Real trigonometric polynomials U, V of degree 128 define exact approximate orbit and kernel functions. Their real cosine/sine coefficients, T, τ and ρ are stored as exact decimal strings in `fold_exact_fourier_inputs_v43.json` and read directly by the validation, not inferred from a later platform FFT. Put $SU(s) = U(s-\sigma)$ and let $A = f_U(U, SU)$, $B = f_{Z_d}(U, SU)$. The orbit and kernel residuals are

$$R = U_s - Tf(U, SU),$$

$$Q = V_s - T\{AV + B\,SV_Z\} - \rho f(U, SU) - (\tau/T)\rho B\,S(U_Z)_s.$$

Here ρ is the period component of the null vector, and B is a column acting on the delayed memory only. The sign of the last term follows by differentiating $U(s-\tau/T)$ with respect to period. The phase constraints are the sine-mode integrals of U_N and V_N. For kernel normalisation, fix its first N cosine coefficient to its nonzero exact baseline value (0.4701250465138572 for the lower fold and 0.1851445547935097 for the upper). This affine normalisation has zero residual by definition, and does not assert nullity of a true continuum operator.

S19.1 Analytic strip bounds

For $u = a_0 + \sum_{j=1}^{m}(a_j \cos 2\pi j s + b_j \sin 2\pi j s)$ on $|\operatorname{Im} s| \le \alpha$, use

$$|u| \le |a_0| + \sum_j (|a_j| + |b_j|)\cosh(2\pi j\alpha),$$

$$|\operatorname{Im} u| \le \sum_j (|a_j| + |b_j|)\sinh(2\pi j\alpha), \quad |u_s| \le \sum_j 2\pi j(|a_j| + |b_j|)\cosh(2\pi j\alpha).$$

Real shifts preserve the strip. For $D = qNE - rN + (r/K)N^2$, these bounds enclose $|D|$ and $|\operatorname{Im} D|$. The test $\kappa|\operatorname{Im} D| < \pi/2$ puts $1 + e^{\kappa D}$ in the open right half-plane, proving analyticity of the principal softplus and bounding the complex sigmoid by one. A positive lower bound on $\operatorname{Re}(1+Z)$ excludes the rational pole. The softplus magnitude is bounded by

$$|\kappa^{-1}\log(1 + e^{\kappa D})| \le \kappa^{-1}\log(1 + e^{\kappa|D|}) + \pi/(2\kappa).$$

Polynomial/rational differentiation then gives component bounds $M_{f,j}$ for f and $M_{h,j}$ for $h = T(AV + B\,SV_Z) + \rho f + (\tau/T)\rho B\,S(U_Z)_s$. The accepted widths are $\alpha = 0.008$ (lower) and 0.006 (upper). All inequalities are evaluated with outward interval arithmetic at 70 decimal digits; coefficient and derivative bounds are recorded in `fold_continuum_stage_v43.json`.

S19.2 Complete residual norm, not just an off-grid sample

For an analytic periodic component bounded by M on the strip, Cauchy's estimate gives $|\widehat{f}_k| \le Me^{-2\pi\alpha|k|}$. A trapezoidal DFT with L equispaced samples aliases only modes $k + \ell L$, so for $|k| < L/2$,

$$\left|\widehat{f}_k - \widehat{f}_k^{\mathrm{DFT}}\right| \le \frac{2Me^{-2\pi\alpha(L-|k|)}}{1 - e^{-2\pi\alpha L}}.$$

The radix-2 interval FFT evaluates the exact-decimal polynomials at $L = 4096$ interval-enclosed roots of unity, evaluates f, h by interval complex arithmetic, and computes interval DFT coefficients. Both evaluation and transform rounding are enclosed; an ordinary floating FFT is not used to certify these coefficients.

With $K_F = 2047$ and $\nu = 1.001 < e^{2\pi\alpha}$, add each finite coefficient modulus, weighted by $\nu^{|k|}$, its alias bound, and the full omitted tail. For example,

$$\|R_j\|_{1,\nu} \le \sum_{|k|\le K_F} \nu^{|k|}\left(|2\pi i k\widehat{U}_{j,k} - T\widehat{f}_{j,k}^{\mathrm{DFT}}| + TA_{j,k}\right) + \frac{2TM_{f,j}(\nu e^{-2\pi\alpha})^{K_F+1}}{1 - \nu e^{-2\pi\alpha}}.$$

The kernel bound has M_h in place of TM_f. The derivative-polynomial coefficients vanish beyond degree 128, so this tail contains every remaining residual mode, including the omitted Nyquist pair. Interval arithmetic uses 65 decimal digits and endpoints are serialised by exact binary-to-decimal outward integer rounding, not nearest formatting.

Fold	Orbit residual upper bounds, N/Z/E	Kernel residual upper bounds, N/Z/E
lower	3.17E-10, 1.88E-11, 5.18E-11	3.97E-12, 4.04E-12, 6.36E-12
upper	2.56E-10, 2.84E-12, 2.13E-11	3.14E-12, 2.70E-12, 1.16E-12

The orbit and kernel phase residuals are respectively at most 1.217×10^{-15} and 2.592×10^{-18} (lower), and 6.978×10^{-15} and 2.376×10^{-17} (upper). These phase-variable norms differ from the physical-time off-grid defects. The exact coefficient strings and componentwise bounds are in `interval_fourier_residual_v43.json`; its source implements the alias/tail proof above.

S19.3 What remains for continuum existence

A residual inclusion is a rigorous statement about an explicitly specified approximate polynomial; it is not a zero inclusion. A complete bordered fold validation additionally needs a controlled inverse including the finite/infinite coupling, a nonlinear contraction or radii inequality, and enclosed continuum transversality/curvature. Varying period/delay changes the shift and can lose derivatives in a naive fixed Wiener-space formulation; a proof must use appropriate smoothing, analytic-strip loss or a regularised formulation. Neither the finite Krawczyk box nor finite Arnoldi multipliers supplies these estimates. The discrepancy of about 5.5×10^{-7} between the upper finite $m = 64$ certificate and the 257-point approximation is therefore important discretisation information, not a contradiction between certificates of the same system. Continuum existence and complete periodic Floquet spectra remain unproved.

S20. Assessment-based dynamic tests and negative endpoints

S20.1 Data, observation gauge and temporal support

ICES SAG stock `her.27.3a47d`, assessment 2025, key 19373 provides total biomass, its reported low/high values, catch, recruitment and age-2–6 fishing pressure. Raw StockDownload, SummaryTable and StockSettings JSON responses, the stock metadata and the cleaned table are archived. The API's total-biomass unit field is blank, whereas catch units are tonnes. Accordingly the proxy

$$N = B/B_{\mathrm{med}}, \quad U = (C/C_{\mathrm{med}})/N$$

is invariant to separate unit rescaling of biomass and catch and is not interpreted as an absolute qE. Reference medians use 1950–1994 only. SSB and reported age-specific fishing pressure are not substituted for total biomass or removal. The incomplete terminal 2025 row is excluded; all test targets are completed years through 2024. The assessment is a single retrospectively revised vintage, not independent annual observations or real-time assessments.

The initial computation used target years 1951–1994 for training but had unequal early target availability across lag candidates. After inspecting its outputs, the support error was corrected by common training targets 1960–1994, giving every candidate observed lagged predictors. Validation remains 1995–2009 and test remains 2010–2024. The initial results are retained under `initial_unequal_support_...`; the post-inspection amendment is explicit in `ANALYSIS_PROTOCOL_v43_P02.md`. Thus this is a transparent retrospective analysis, not external preregistration or an untouched confirmatory holdout. Secondary ecological/recursive diagnostics were specified after that initial response test and before their own estimation and remain exploratory.

S20.2 Response laws and selection

For $D_y = N_{y-1} - N_y$, memory is

$$Z_y = e^{-1/t_m} Z_{y-1} + (1 - e^{-1/t_m}) \log(1 + e^{10 D_y})/10,$$

initialised in 1947 at $\log(2)/10$. Origin year y predicts U_{y+1} with $Z_{y-\ell}$ held throughout the one-year scalar flow. The mobilising law is

$$\dot{U} = (1 - U/M)\{\eta U(Z - U/M) + d_0 Z/(1 + Z)\};$$

the protective law is

$$\dot{U} = (1 - U/M)\eta_p\{fM/(1 + Z) - U\}, \quad 0 < f < 1.$$

Both have two continuously fitted coefficients. Candidate memory times are $\{0.25, 1, 2, 5, 10\}$, integer lags $0, \ldots, 12$, and cap multipliers $\{1.05, 1.5, 2, 4\}$, giving 260 candidates per law. The cap is the fixed multiplier times the maximum origin proxy in the fitting period; its deterministic reference maximum is recomputed for the pretest refit, not selected using the test. Training least squares estimates the two coefficients; validation RMSE selects the discrete candidate; the two coefficients are refitted on all 50 pretest targets. Test performance never selects a hyperparameter. Bounded local least squares starts from nonnegative least-squares fits of the first-order rate approximation; it does not certify a global parameter optimum. Response rates are bounded above by 20, with positive lower bounds of 10^{-7} for adjustment rates and 10^{-8} for the mobilising additive coefficient; the protective cap fraction lies in $[10^{-6}, 0.999999]$. Full solver tolerances and candidate diagnostics are in the executable source and grid.

The common-support selection is $(t_m, \ell, M/\max U) = (0.25, 5, 4)$ for mobilising and $(0.25, 2, 1.05)$ for protective. The fitted lags are tuning values, not estimates of deployment delay. Scalar RK4 uses 64 substeps/year and is checked at 128; maximum central prediction discrepancies are below 6×10^{-13}. Persistence and affine AR(1) use exactly the same target support. Catch intensity derives partly from the stock variable driving the signal, and stock/catch share an age-structured assessment; neither predictor independence nor direct observation of effort is assumed.

Biomass scenario	Response	RMSE	RMSE/persistence	RMSE/AR(1)
TBiomass	mobilising	0.11634112	0.89973897	1.03523325
TBiomass	protective	0.11185499	0.86504488	0.99531448
Low_TBiomass	mobilising	0.13302539	0.89572104	1.02850383
Low_TBiomass	protective	0.12920732	0.87001218	0.99898385
High_TBiomass	mobilising	0.10175958	0.90072260	1.03688734
High_TBiomass	protective	0.09721446	0.86049160	0.99057450

The mobilising additive coefficient is near its positive lower bound 10^{-8}; this is not an interior well-identified fit. The protective central RMSE is about 13.5% below persistence but only 0.47% below AR(1), on 15 dependent annual targets. The low/high biomass paths are coherent stress scenarios, not independent observations, a simultaneous confidence set or replication. They cannot establish a robust causal mechanism from such a small gain.

S20.3 Ecological competitor and recursive sampled surrogate

A secondary observed-input ecological model has exact held-input annual flow

$$\dot{N} = (r - hU)N - bN^2 + \beta R_{y-g}, \quad r, b, h > 0, \quad \beta \geq 0,$$

where R is recruitment normalised by its 1960–1994 median. The logistic model sets $\beta = 0$; the competing augmented model selects $g \in \{1, 2, 3\}$ by validation and refits on pretest targets. Coefficient h absorbs the observation gauge; it is not assumed to be a measured absolute removal coefficient. The exact Riccati solution, not an Euler stock step, is fitted by least squares. Lag-one recruitment is selected, reducing test error markedly, but the resulting stock prediction still fails both baselines. Neither the scalar-logistic ecology nor an institutional interpretation is rescued by the response-only fit.

The exploratory recursive annual surrogate starts from the observed 2009 N, U and pre-2010 memory history. With logistic parameters fitted on pretest data, each future step (i) flows stock for one year with U_y held, (ii) updates the scalar effort command from start-year delayed memory using its exact-to-numerical-accuracy scalar flow, and (iii) updates memory from the modelled stock decline. No future observed stock, catch or recruitment enters. This particular causal sampled surrogate is not claimed identical to the continuous DDE or the flow-then-end-review operator in the local theorem. Its comparator is recursively iterated affine VAR(1) fitted to (N, U) on the same pretest targets; persistence is the

constant 2009 state, not annually refreshed persistence. Componentwise one-step and recursive metrics are therefore distinct endpoints.

Stock component	RMSE	RMSE/persistence	RMSE/AR(1)
logistic	0.10812702	1.35472544	1.35467534
recruitment_lag1	0.08009500	1.00351172	1.00347461

Recursive law	Stock/persistence	Stock/VAR(1)	Proxy/persistence	Proxy/VAR(1)
mobilising	2.35468748	3.16605824	0.35709304	1.25258336
protective	1.93479164	2.60147602	0.22181273	0.77805756

Both recursive stock paths fail persistence and VAR(1). The protective proxy path beats both, but cannot validate a joint mechanism whose stock path fails. The fitted protective global sufficient gain is approximately 2.31309, outside Theorem 7.5; exceeding one is not a proof of instability. The source scripts, complete grids, pointwise predictions, domain checks and raw-data hash are supplied. No uncertainty envelope is presented as repeated field evidence, and no theoretical delay threshold is transported to herring. The result is a reproducible field challenge to the scalar ecology and a small conditional response advantage, not full dynamic field validation.